

National University of Computer and Emerging Sciences



Practice Quiz (Ungraded)
for
Object Oriented Programming Lab

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Ungraded Quiz

Quiz Duration: 60 minutes

Instructions:

- Read the questions carefully.
- Since majority voted for practice quiz, we are having this quiz
- Attendance of lab 8 will be marked on the basis of submission of this Quiz
- Attempt this Quiz after preparing till lab 8
- This quiz is only for practice, so please attempt it honestly. The goal is to help you build confidence before the midterm exams.
- If you find the quiz difficult, do not feel discouraged. Instead, use it as motivation to practice more before the mids. If you have any confusion, feel free to ask questions through email or Google Chat so your concepts can be clarified.
- It is highly recommended that everyone attempts the quiz.
- However, if someone does not want to attempt the full quiz, you should at least attempt a small portion of it and submit it. This will ensure that your attendance can still be marked.

Concepts covered

- Cascading setters
- Static members
- Copy constructor
- Deep copy using char*
- Pointer-based memory handling
- Static member functions
- Class design

Backend Developer Task – AJ Islamic Library

Your lab teacher **Ahmad Jawad** started a YouTube channel in collaboration with his school friend **Hafiz Arham**, whose **Qur'an recitation is very beautiful**. Your teacher manages the channel by **editing and organizing the videos**.

Suppose he has hired you as a **junior backend developer** to extend a small **team management system** for the YouTube channel **AJ Islamic Library**.

Your task is to design a **C++ class to manage team members and track recitation progress of Quran Paras (1–30)**.

Class Requirements

Create a class named:

- TeamMember

Private Data Members

- char* memberName
- char* role
- bool reciter

Static Members

- static int totalMembers
- static bool paraCompleted[30]

Explanation

- totalMembers → counts the total number of team members created.
- paraCompleted[30] → tracks which **Quran paras have been recited and recorded**.

Mapping example:

- index 0 → para 1
- index 1 → para 2
- ...
- index 29 → para 30

Role Restrictions

Only the following roles are allowed:

- founder
- co founder
- co founder and reciter
- reciter
- content writer
- editor
- manager

If any other role is entered, it should **not be accepted**.

Automatic Reciter Detection

The variable:

- bool reciter

must NOT be taken from the user.

It should automatically become **true** if the role contains:

Reciter (should be set true for “co founder and reciter” or “reciter”)

Otherwise it should be **false**.

Constructors

You must implement:

1. **Default constructor**
2. **Overloaded constructor**
3. **Copy constructor**

Deep copy must be applied to:

- memberName
- role

Cascading Setters

Implement cascading setters for:

- `setMemberName(char*)`
- `setRole(char*)`

Example usage:

```
member.setMemberName("Ahmad Jawad").setRole("founder");
```

Static Functions

- **`markParaCompleted(int paraNumber)`**

Marks the recitation progress.

- **`displayProgress()`**

Display:

Paras Completed:

Paras In Progress:

Display Function

Create a function:

`display()`

Output format:

Member Name:

Role:

Reciter: Yes / No

Total Members in AJ Islamic Team:

Main Function Requirements

Create objects to represent the team.

1. Ahmad Jawad

Create using **default constructor**, then use setters.

Ahmad Jawad

founder

2. Hafiz Arham

Create using **overloaded constructor**.

hafiz arham

co founder and reciter

3. Tahir

Default constructor + setters.

tahir

content writer

4. Waleed

Create using **copy constructor from Tahir**, then update the name.

waleed

5. Hafiz Abdulrehman

Create using **overloaded constructor**.

H M Aly Khan

reciter

Final Program Steps

1. Display all team members.
2. Mark para 9, 28, 29 and 30 as completed using markParaCompleted().
3. Call displayProgress().

----- Happy Coding! 😊 -----

If you want to keep enjoying Quran recitation videos and short reminders, feel free to subscribe to
AJ Islamic Library on YouTube!